

CLIMATE CLASSIFICATION (KOPPEN) / INDIA CLIMATIC REGIONS

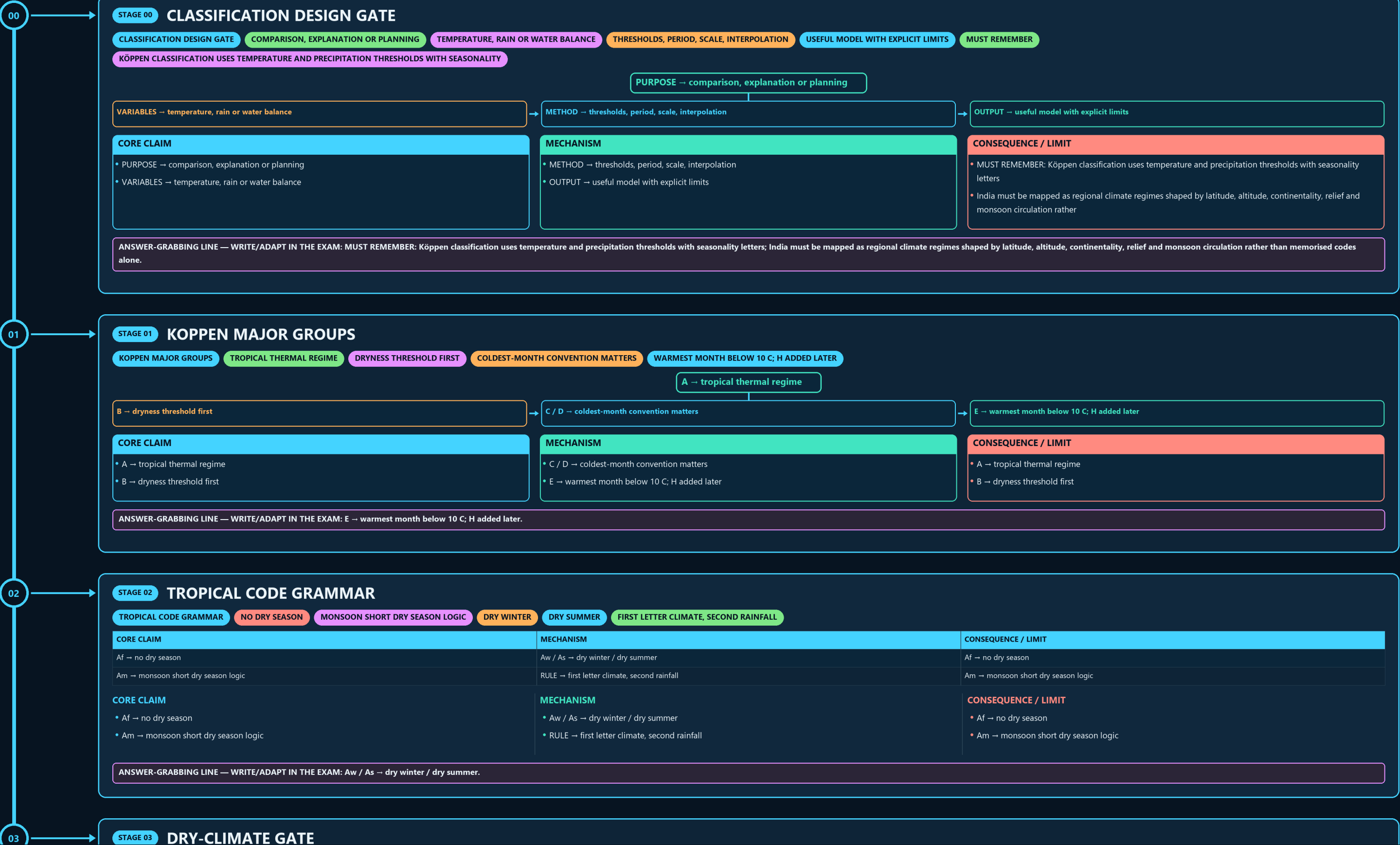
CLASSIFICATION DESIGN GATE → KOPPEN MAJOR GROUPS → TROPICAL CODE GRAMMAR → DRY-CLIMATE GATE → C-D THRESHOLD FIREWALL → NORMAL-TO-TREND LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



Af → no dry season

Am → monsoon short dry season logic

Aw / As → dry winter / dry summer

RULE → first letter climate, second rainfall

Af → no dry season

Am → monsoon short dry season logic

CORE CLAIM

- Af → no dry season
- Am → monsoon short dry season logic

MECHANISM

- Aw / As → dry winter / dry summer
- RULE → first letter climate, second rainfall

CONSEQUENCE / LIMIT

- Af → no dry season
- Am → monsoon short dry season logic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Aw / As → dry winter / dry summer.

03

STAGE 03

DRY-CLIMATE GATE

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TEMPERATURE + RAIN SEASON

DRYNESS THRESHOLD

PRECIPITATION BELOW THRESHOLD

B GROUP

BW DESERT

LESS DRY

BS STEPPE; H/K THERMAL SUFFIX

TEMPERATURE + RAIN SEASON → dryness threshold

PRECIPITATION BELOW THRESHOLD → B group

DRIER → BW desert

LESS DRY → BS steppe; h/k thermal suffix

CORE CLAIM

- TEMPERATURE + RAIN SEASON → dryness threshold
- PRECIPITATION BELOW THRESHOLD → B group

MECHANISM

- DRIER → BW desert
- LESS DRY → BS steppe; h/k thermal suffix

CONSEQUENCE / LIMIT

- TEMPERATURE + RAIN SEASON → dryness threshold
- PRECIPITATION BELOW THRESHOLD → B group

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LESS DRY → BS steppe; h/k thermal suffix.

04

STAGE 04

C-D THRESHOLD FIREWALL

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COLDEST MONTH

C/D DISCRIMINATOR

CONVENTION 1

MINUS 3 C BOUNDARY

CONVENTION 2

0 C BOUNDARY

ANSWER RULE

CORE CLAIM

COLDEST MONTH → C/D discriminator

CONVENTION 1 → minus 3 C boundary

MECHANISM

CONVENTION 2 → 0 C boundary

ANSWER RULE → state convention before mapping

CONSEQUENCE / LIMIT

COLDEST MONTH → C/D discriminator

CONVENTION 1 → minus 3 C boundary

CORE CLAIM

- COLDEST MONTH → C/D discriminator
- CONVENTION 1 → minus 3 C boundary

MECHANISM

- CONVENTION 2 → 0 C boundary
- ANSWER RULE → state convention before mapping

CONSEQUENCE / LIMIT

- COLDEST MONTH → C/D discriminator
- CONVENTION 1 → minus 3 C boundary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONVENTION 2 → 0 C boundary; ANSWER RULE → state convention before mapping.

05

STAGE 05

NORMAL-TO-TREND LADDER

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STATED 30-YEAR REFERENCE

OBSERVATION MINUS NORMAL

SPREAD AND RECURRENCE

MULTI-PERIOD DIRECTION WITH UNCERTAINTY

NORMAL → stated 30-year reference

ANOMALY → observation minus normal

VARIABILITY → spread and recurrence

TREND → multi-period direction with uncertainty

CORE CLAIM

- NORMAL → stated 30-year reference
- ANOMALY → observation minus normal

MECHANISM

- VARIABILITY → spread and recurrence
- TREND → multi-period direction with uncertainty

CONSEQUENCE / LIMIT

- NORMAL → stated 30-year reference
- ANOMALY → observation minus normal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VARIABILITY → spread and recurrence; TREND → multi-period direction with uncertainty.

06

STAGE 06

CLIMATE-BOUNDARY MODEL

CLIMATE-BOUNDARY MODEL

MEASURED VARIABLES

QUALITY CONTROL

COMPARABLE RECORDS

INTERPOLATION + THRESHOLDS

CLASS MAP

GROUND REALITY

TRANSITION, NOT WALL

STATIONS / GRIDS → measured variables

QUALITY CONTROL → comparable records

INTERPOLATION + THRESHOLDS → class map

GROUND REALITY → transition, not wall

- NORMAL → stated 50-year reference
- ANOMALY → observation minus normal

- CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
- climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

09

STAGE 09

CLASSIFICATION COMPARISON

CLASSIFICATION COMPARISON

TEMPERATURE + PRECIPITATION THRESHOLDS

MODIFIED THERMAL ZONATION

WATER BALANCE + PET

MATCH METHOD TO QUESTION PURPOSE

CLOSE DISTINCTION

A/B/C/D/E ARE MAJOR CLIMATE GROUPS, SECOND AND THIRD LETTERS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET	CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose	climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

CORE CLAIM

- KOPPEN → temperature + precipitation thresholds
- TREWARTHA → modified thermal zonation

MECHANISM

- THORNTHWAITE → water balance + PET
- CHOICE → match method to question purpose

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
- climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

10

STAGE 10

PLANNING OVERLAY

PLANNING OVERLAY

CLIMATE TYPE

LONG-PERIOD THERMAL-RAIN PATTERN

CROP AND SEASON CONTEXT

SOIL + LANDFORM + GROWING PERIOD

RISK LAYER

HAZARDS, EXPOSURE AND VULNERABILITY

EVIDENCE LIMIT

CLIMATE TYPE → long-period thermal-rain pattern

AGRO-CLIMATE → crop and season context

→

AGRO-ECOLOGY → soil + landform + growing period

→

RISK LAYER → hazards, exposure and vulnerability

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
- CLIMATE TYPE → long-period thermal-rain pattern - AGRO-CLIMATE → crop and season context	- AGRO-ECOLOGY → soil + landform + growing period - RISK LAYER → hazards, exposure and vulnerability	- EVIDENCE LIMIT: Threshold classification simplifies transitional and highland mosaics - map answers should state dataset/scale and avoid treating a code boundary as a permanent administrative or ecological line.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Threshold classification simplifies transitional and highland mosaics; map answers should state dataset/scale and avoid treating a code boundary as a permanent administrative or ecological line.

11

STAGE 11

CLIMATE ANSWER SPINE

CLIMATE ANSWER SPINE

PURPOSE, VARIABLES AND PERIOD

LETTERS AND THRESHOLDS

INDIA WITH TRANSITION CAVEAT

METHOD, USE, LIMITATION AND VERDICT

DEFINE → purpose, variables and period

→

DECODE → letters and thresholds

→

MAP → India with transition caveat

→

COMPARE → method, use, limitation and verdict

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
- DEFINE → purpose, variables and period - DECODE → letters and thresholds	- MAP → India with transition caveat - COMPARE → method, use, limitation and verdict	- DEFINE → purpose, variables and period - DECODE → letters and thresholds

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP → India with transition caveat; COMPARE → method, use, limitation and verdict.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

A CLIMATES IN INDIA INCLUDE TROPICAL HUMID CLIMATES; AM

B CLIMATES ARE DRY

BSHW SEMI-ARID STEPPE AND BWHW HOT DESERT.

CWG IS THE GANGES/NORTH INDIAN PLAINS MONSOON CLIMATE WITH

H/E APPLIES TO THE HIMALAYA ABOVE HIGH ELEVATIONS/SNOW-LINE CONDITIONS.

TREWARTHA MODIFIES KOPPEN AND INCLUDES HIGHLAND CLIMATE FOR INDIA.

THORNTHWAITE USES PRECIPITATION–EVAPOTRANSPIRATION MOISTURE BALANCE, NOT TEMPERATURE ALONE.

OPTIONAL SOURCE DEPTH	ADVANCED DEBATE	LEGACY / NUANCE
- A climates in India include tropical humid climates; Am relates to heavy monsoon rain, while Aw marks savanna/dry-winter interior. - B climates are dry: BShw semi-arid steppe and BWhw hot desert. - Cwg is the Ganges/north Indian plains monsoon climate with dry winter. - H/E applies to the Himalaya above high elevations/snow-line conditions.	- Trewartha modifies Koppen and includes highland climate for India. - Thorntwaite uses precipitation–evapotranspiration moisture balance, not temperature alone. - India has Koppen A, B, C and H in standard applied schemes. - Am/Amw = West Coast and very wet monsoon belt; Aw = peninsular savanna interior.	- As = Tamil Nadu/Coromandel dry-summer tropical type linked to NE monsoon rainfall. - BWhw = hot desert of western Rajasthan/Thar/Kachchh. - BShw = semi-arid steppe/rain-shadow and NW margins. - Cwg = Ganges type: north Indian plains, monsoon with dry winter.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.